Lecture 4: Convolution, Causality, and Stability

CSE 219: Signals and Linear Systems

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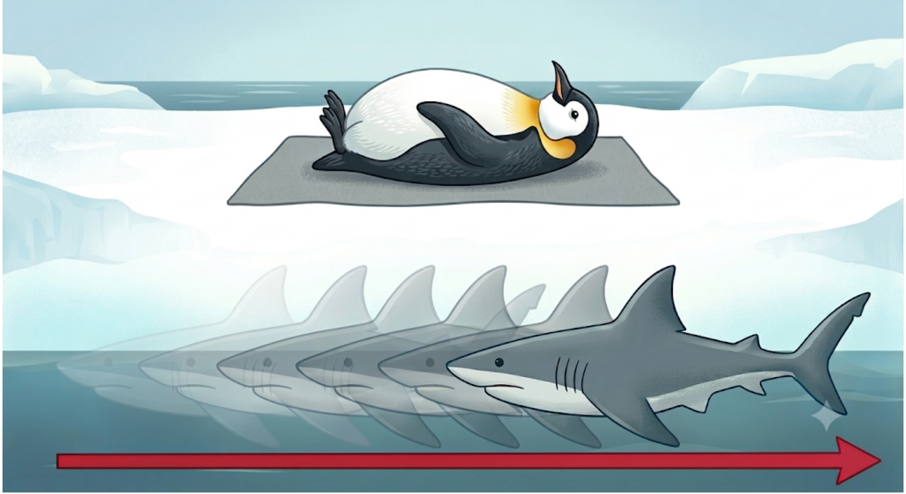
Recall: Keep the first signal just as is, flip the second, slide rightward, find overlaps, then multiply-and-add!



Recall: discrete-time convolution is sliding multiply-and-add

For two discrete-time signals $x[n]$ and $h[n]$, their convolution is

$$\boxed{y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k].}$$

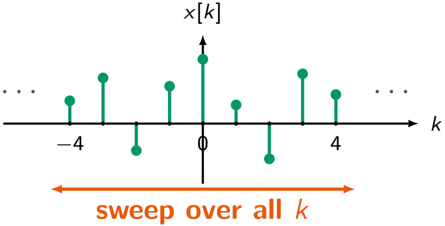
What the formula is saying

$$x[k] \xrightarrow{\text{multiply}} h[n-k] \xrightarrow{\text{add over } k} y[n]$$

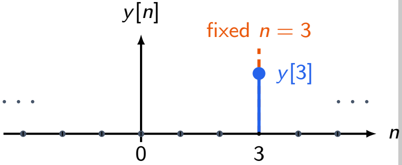
For each fixed n, the sum produces one output sample $y[n]$.

Convolution gives a whole output signal, one value at a time

Sweep over the input index k

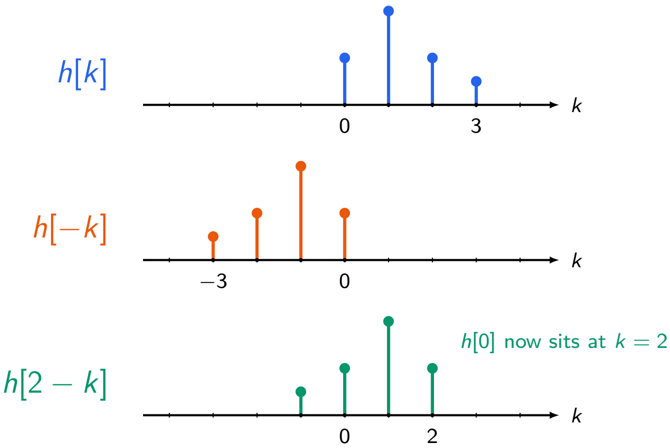


Fix one output index



$$y[3] = \sum_{k=-\infty}^{\infty} x[k]\, h[3 - k]$$

For fixed n, $h[n-k]$ is made by reflect-then-right-shift



The continuous-time version replaces the sum by an integral

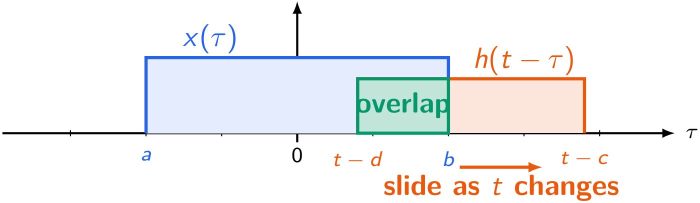
For continuous-time signals, the idea is the same:
reflect, shift, multiply, and add continuously.

Continuous-time convolution

$$\boxed{y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau}$$

For a fixed t, sweep along τ and integrate along the overlap

Fix t, then slide $h(t - \tau)$ along the τ-axis



For this one fixed t,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

Discrete and continuous convolution follow the same recipe

1

Choose the output location.

For discrete time, choose n. For continuous time, choose t.

2

Reflect and shift the second signal.

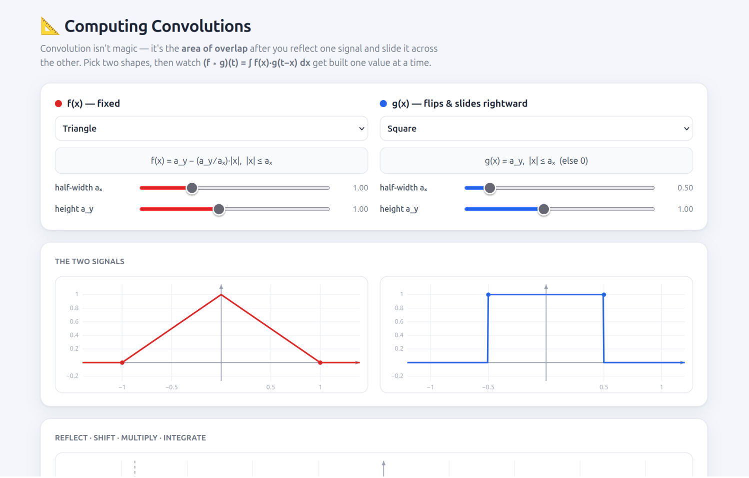
Discrete: $h[k] \to h[-k] \to h[n-k]$. Continuous: $h(\tau) \to h(-\tau) \to h(t-\tau)$.

3

Multiply where they overlap, then add.

Discrete: sum over k. Continuous: integrate over τ.

Please try this demo on computing convolutions!



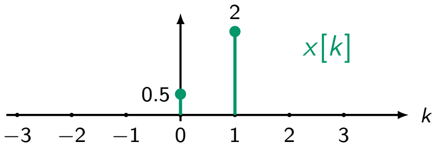
Practice Problems on Convolution

Practice (Example 2.2)

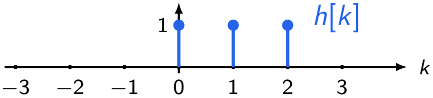
For the two signals below, compute

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

Input $x[k]$

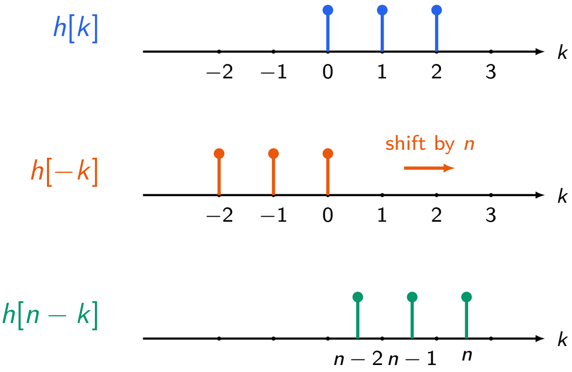


Impulse response $h[k]$



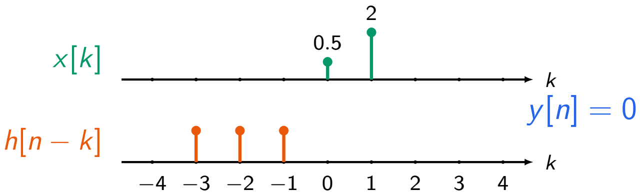
Answer setup: $h[n-k]$ is the reflected signal shifted by n

For each fixed n, draw $h[n-k]$ as a function of k.



Answer: start sliding $h[n-k]$ past $x[k]$

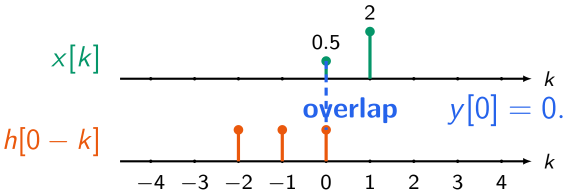
$n < 0$: **no overlap**



For $n < 0$, the two signals do not share any nonzero overlap of k

Answer: start sliding $h[n-k]$ past $x[k]$

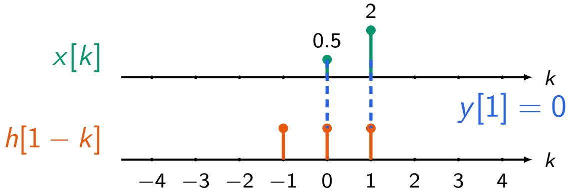
$n = 0$: **one overlap**



$y[0] = 0.5 \times 1 = 0.5$

Answer: when the signals overlap more, the sum grows

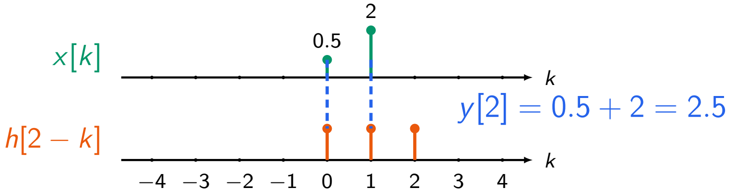
$n = 1$: **two overlaps**



$y[1] = 0.5 + 2 = 2.5$

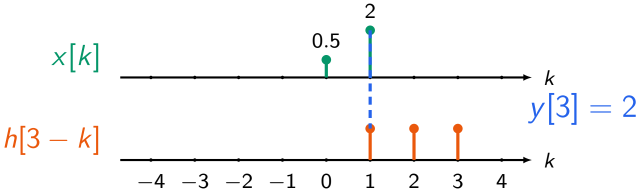
Answer: when the signals overlap more, the sum grows

$n = 2$: **still two overlaps**



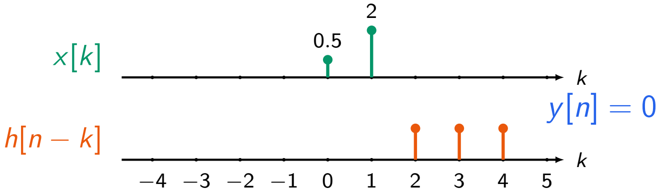
Answer: after the last overlap, the output returns to zero

$n = 3$: **one overlap**



Answer: after the last overlap, the output returns to zero

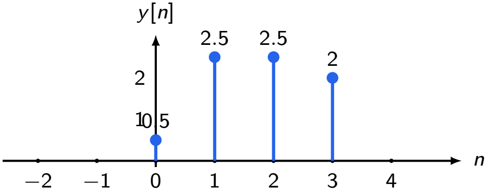
$n > 3$: no overlap again



After $n = 3$, the sliding signal has passed the support of $x[k]$

Answer: final convolution output

$$y[n] = \begin{cases} 0.5, & n = 0, \\ 2.5, & n = 1, \\ 2.5, & n = 2, \\ 2, & n = 3, \\ 0, & \text{otherwise.} \end{cases}$$

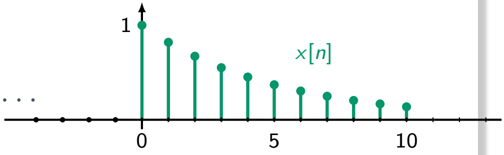


Practice (Example 2.3)

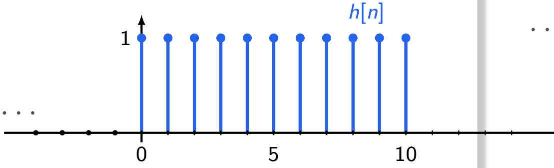
Let $0 < \alpha < 1$. Compute

$$y[n] = x[n] * h[n], \qquad x[n] = \alpha^n u[n], \qquad h[n] = u[n].$$

Input $x[n] = \alpha^n u[n]$

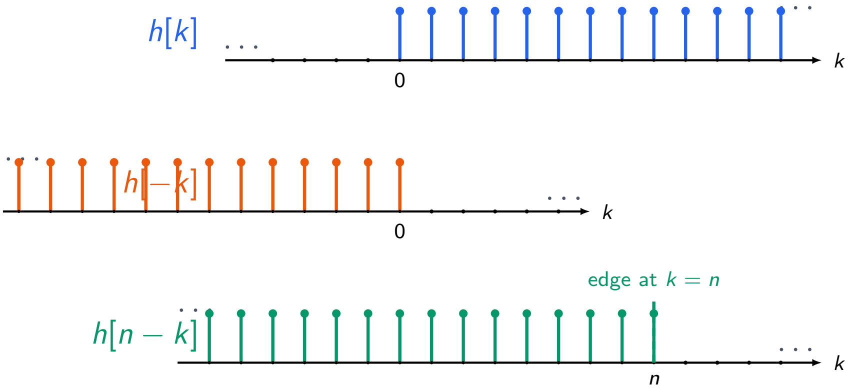


Impulse response $h[n] = u[n]$

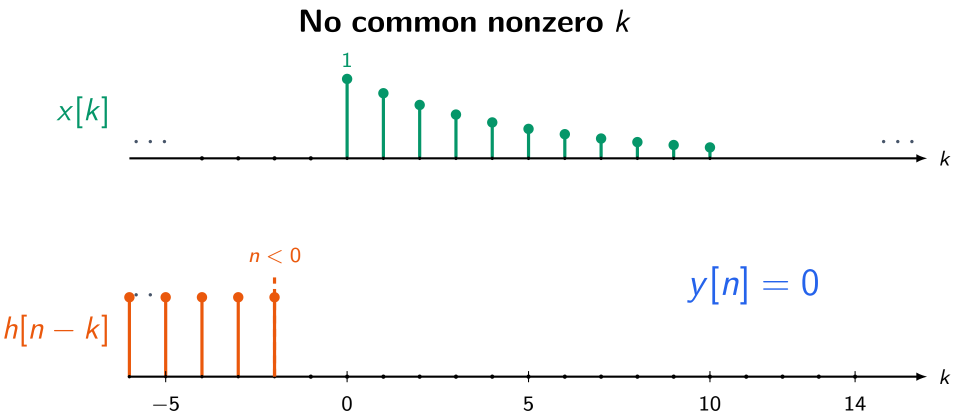


First understand how $h[n-k]$ looks like

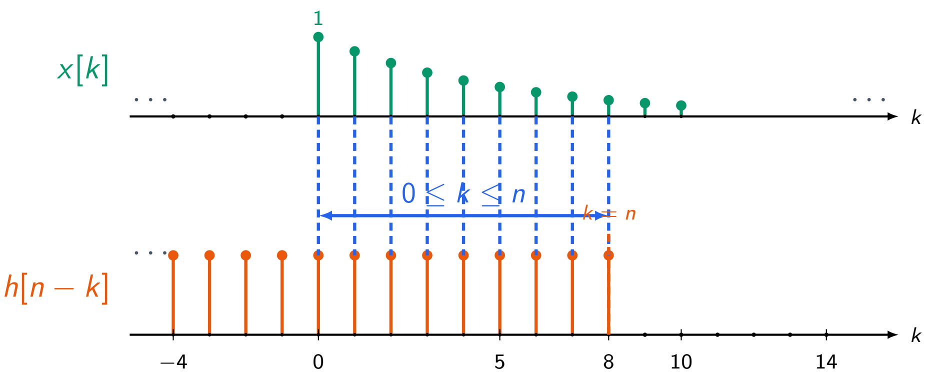
Reflect $h[k]$, then shift it by n



For $n < 0$, there is no overlap



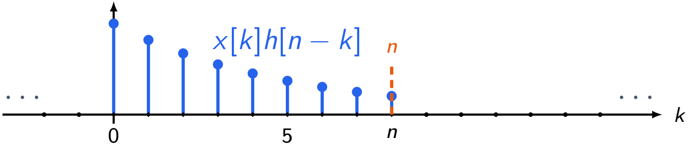
For $n \geq 0$, the overlap is from $k = 0$ to $k = n$



$$\text{Overlap} \quad \Rightarrow \quad x[k]h[n-k] = \alpha^k, \qquad 0 \leq k \leq n.$$

Therefore the product is just the overlapping part of $x[k]$

Product sequence for $n \geq 0$



$$x[k]h[n-k] = \begin{cases} \alpha^k, & 0 \leq k \leq n, \\ 0, & \text{otherwise.} \end{cases}$$

Now sum the product over all k

Only the overlap region contributes to the convolution sum.

$$\textbf{For } n \geq 0$$

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\ &= \sum_{k=0}^{n} \alpha^k \\ &= 1 + \alpha + \alpha^2 + \cdots + \alpha^n. \end{aligned}$$

The finite geometric sum gives the final answer

For $0 < \alpha < 1$, $1 + \alpha + \alpha^2 + \cdots + \alpha^n = \frac{1-\alpha^{n+1}}{1-\alpha}$.

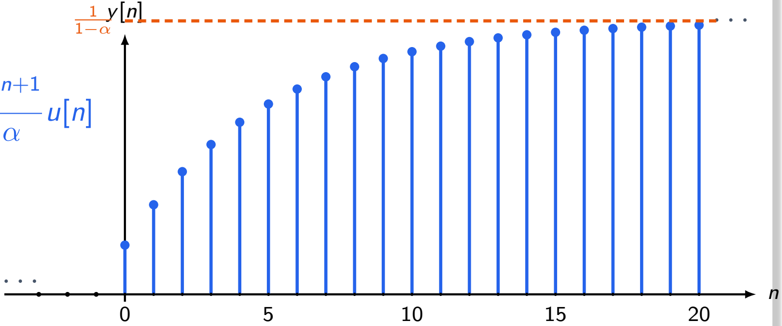
Final convolution output

$$y[n] = \begin{cases} \dfrac{1-\alpha^{n+1}}{1-\alpha}, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

The output rises and approaches a limiting value

Shape of $y[n]$

$$y[n] = \frac{1-\alpha^{n+1}}{1-\alpha}u[n]$$

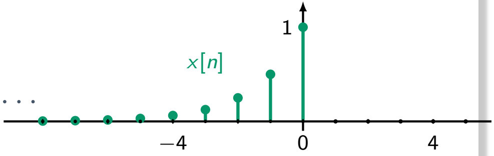


Practice (Example 2.5)

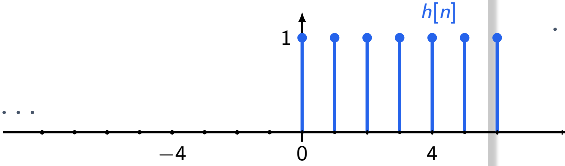
Compute

$$y[n] = x[n] * h[n], \qquad x[n] = 2^n u[-n], \qquad h[n] = u[n].$$

Input $x[n] = 2^n u[-n]$

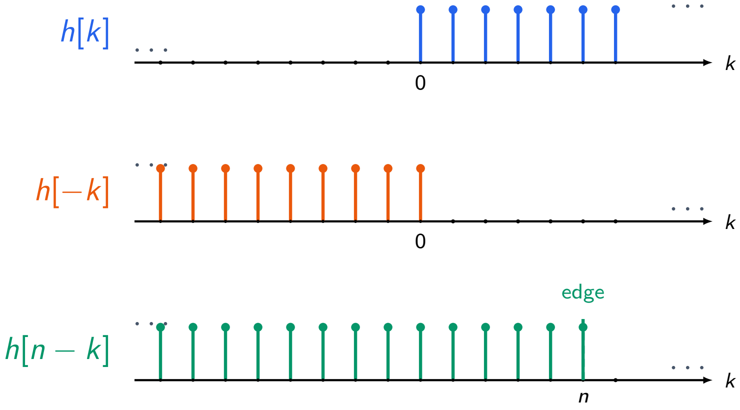


Impulse response $h[n] = u[n]$

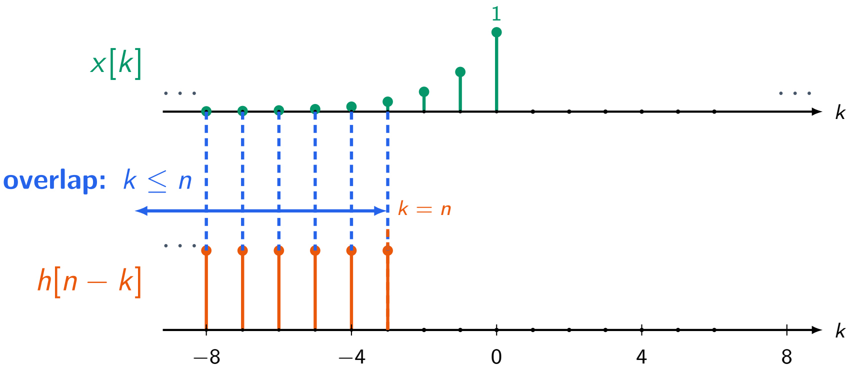


First understand how $h[n-k]$ looks

Reflect $h[k]$, then shift it by n

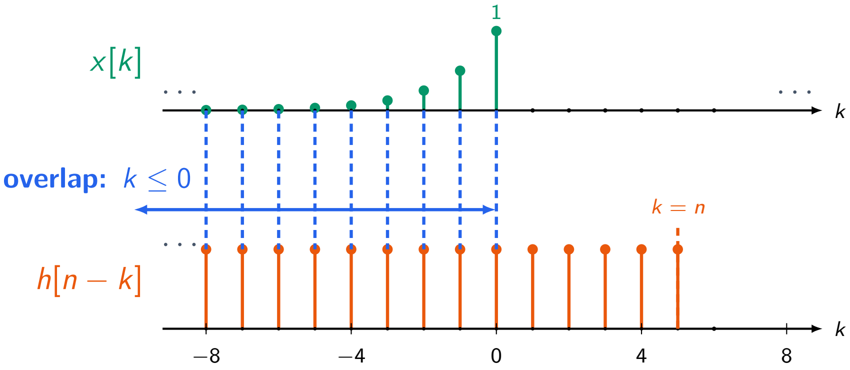


For $n < 0$, the overlap stops at $k = n$



For $n < 0, \quad x[k]h[n-k] = 2^k$ for $k \leq n$.

For $n \geq 0$, all nonzero samples of $x[k]$ overlap



For $n \geq 0$, $\quad x[k]h[n-k] = 2^k$ for $k \leq 0$.

For $n < 0$, the sum stops at $k = n$

Case 1: $n \quad < \quad 0$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$
$$= \sum_{k=-\infty}^{n} 2^k.$$

$$\sum_{k=-\infty}^{n} 2^k = 2^n + 2^{n-1} + 2^{n-2} + \cdots = 2^{n+1}.$$

For $n \geq 0$, the sum includes all left-sided samples

Case 1: $n \geq 0$

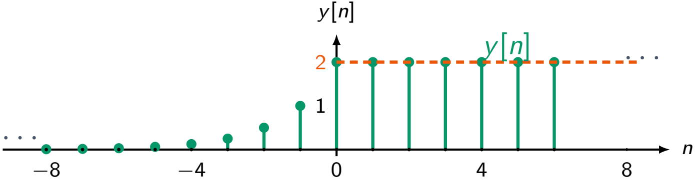
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$
$$= \sum_{k=-\infty}^{0} 2^k.$$

$$\sum_{k=-\infty}^{0} 2^k = \cdots + \frac{1}{8} + \frac{1}{4} + \frac{1}{2} + 1 = 2.$$

The output is left-sided exponential, then a constant

Final answer

$$y[n] = \begin{cases} 2^{n+1}, & n < 0, \\ 2, & n \geq 0. \end{cases}$$

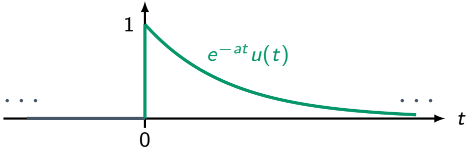


Practice (Example 2.6)

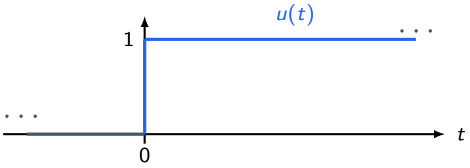
Let $a > 0$. Compute

$$y(t) = x(t) * h(t), \qquad x(t) = e^{-at}u(t), \qquad h(t) = u(t).$$

Input $x(t) = e^{-at}u(t)$

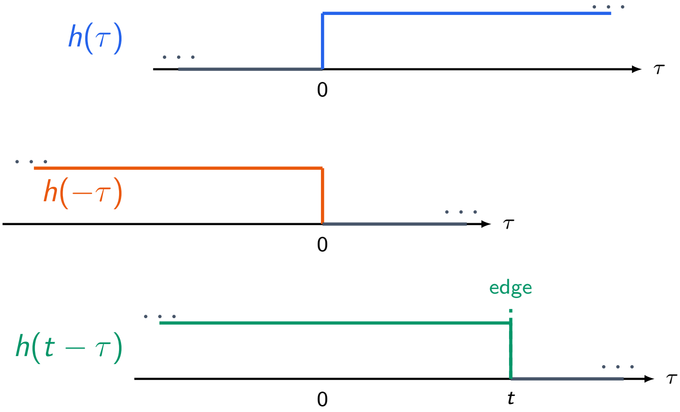


Impulse response $h(t) = u(t)$

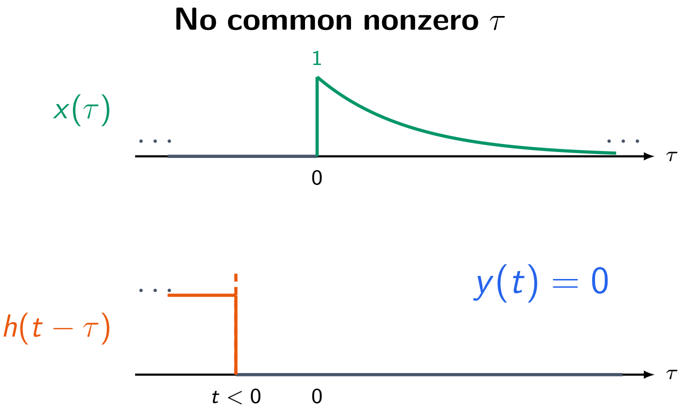


First understand how $h(t-\tau)$ moves

Reflect $h(\tau)$, then shift by t

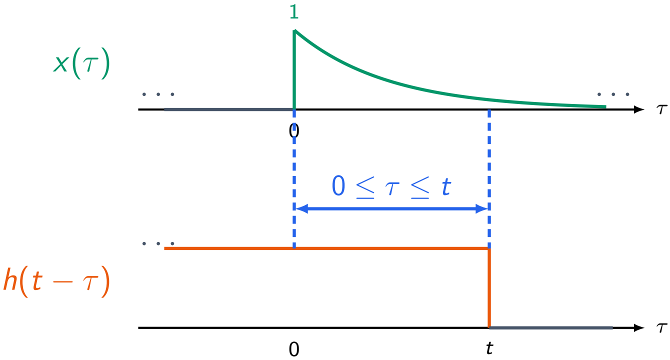


For $t < 0$, there is no overlap



No overlap $\Rightarrow$ $y(t) = 0.$

For $t > 0$, the overlap is $0 \leq \tau \leq t$



Overlap $\Rightarrow$ $x(\tau)h(t-\tau) = e^{-a\tau}$, $0 \leq \tau \leq t$.

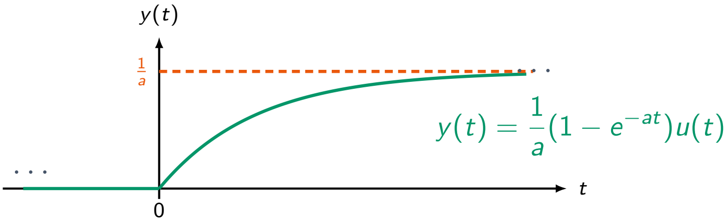
Now integrate only over the overlap

$$
\begin{aligned}
\text{For } t > 0,\ y(t) &= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau \\
&= \int_0^t e^{-a\tau}\,d\tau \\
&= \left[-\frac{1}{a}e^{-a\tau}\right]_0^t \\
&= \frac{1}{a}\left(1 - e^{-at}\right).
\end{aligned}
$$

For $t < 0$, there is no overlap, so $y(t) = 0$.

The output gradually rises to $1/a$

$$y(t) = \frac{1}{a}\left(1 - e^{-at}\right) u(t).$$



As t increases, the overlap grows, so the integral rises toward $\dfrac{1}{a}$.

Property: Convolution distributes over addition

If one signal is a sum of two simpler signals,
we may convolve the two parts separately.

$$x[n] * \big(h_1[n] + h_2[n]\big) = x[n] * h_1[n] + x[n] * h_2[n].$$

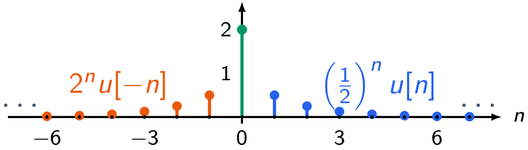
$$\big(x_1[n] + x_2[n]\big) * h[n] = x_1[n] * h[n] + x_2[n] * h[n].$$

Practice (Example 2.10)

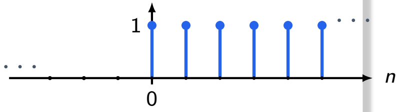
Compute

$$y[n] = x[n] * h[n], \qquad x[n] = \left(\frac{1}{2}\right)^n u[n] + 2^n u[-n], \qquad h[n] = u[n].$$

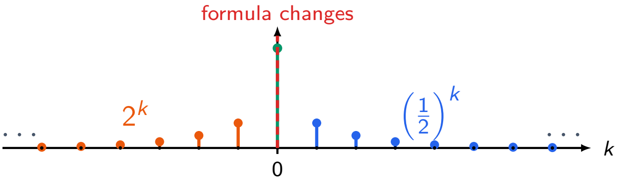
Input $x[n]$



$h[n] = u[n]$



Oops, different input expressions on the left and right side!



$$x[k] = \underbrace{2^k u[-k]}_{\text{left-sided part}} + \underbrace{\left(\frac{1}{2}\right)^k u[k]}_{\text{right-sided part}} .$$

So, we split it first.

Split the problem into two easier ones!

Split $x[n]$, then convolve separately

$$x[n] = x_1[n] + x_2[n], \qquad x_1[n] = \left(\frac{1}{2}\right)^n u[n], \qquad x_2[n] = 2^n u[-n].$$

$$\begin{aligned} y[n] &= x[n] * h[n] \\ &= x_1[n] * h[n] + x_2[n] * h[n] \\ &= y_1[n] + y_2[n]. \end{aligned}$$

Each part is one we already know how to do.

The two smaller convolutions have simple answers

Left-sided part

$$x_2[n] = 2^n u[-n], \qquad h[n] = u[n].$$

$$y_2[n] = x_2[n] * h[n].$$

$$y_2[n] = \begin{cases} 2^{n+1}, & n < 0, \\ 2, & n \geq 0. \end{cases}$$

Right-sided part

$$x_1[n] = \left(\frac{1}{2}\right)^n u[n], \qquad h[n] = u[n].$$

$$y_1[n] = x_1[n] * h[n].$$

$$y_1[n] = \left(2 - 2^{-n}\right) u[n].$$

Now add the two outputs

Final expression

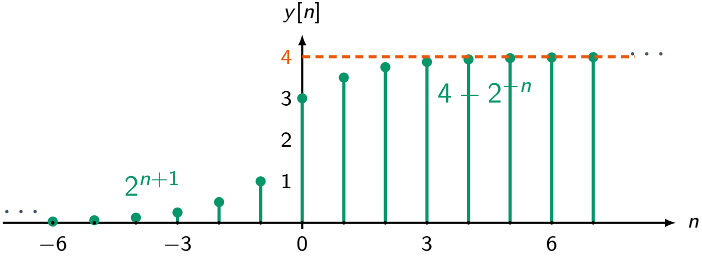
$$y[n] = y_1[n] + y_2[n].$$

$$\boxed{y[n] = \begin{cases} 2^{n+1}, & n < 0, \\ 4 - 2^{-n}, & n \geq 0. \end{cases}}$$

The left side rises toward 1, then the right side rises toward 4.

The output approaches 4 on the right

Sketch of $y[n]$



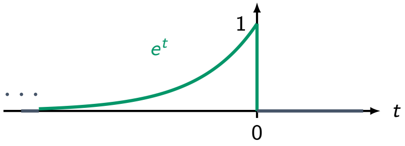
$$y[n] = \begin{cases} 2^{n+1}, & n < 0, \\ 4 - 2^{-n}, & n \geq 0. \end{cases}$$

Practice: convolve two one-sided exponentials

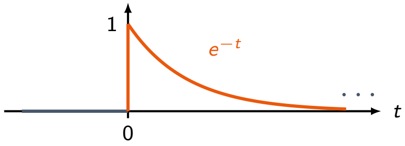
Find $y(t) = x(t) * h(t)$, where

$$x(t) = e^t u(-t), \qquad h(t) = e^{-t} u(t).$$

$$x(t) = e^t u(-t)$$



$$h(t) = e^{-t} u(t)$$



Answer: here the second signal is not just a step

First find the overlap

$$x(\tau) = e^{\tau} u(-\tau) \quad \Longrightarrow \quad x(\tau) \neq 0 \text{ only for } \tau \leq 0.$$

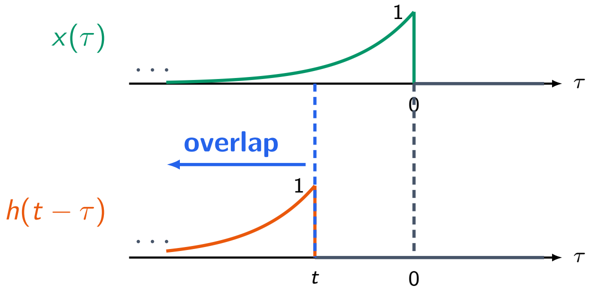
$$h(t-\tau) = e^{-(t-\tau)} u(t-\tau) \quad \Longrightarrow \quad h(t-\tau) \neq 0 \text{ only for } \tau \leq t.$$

Therefore the overlap is $\tau \leq \min(0, t)$. On the overlap,

$$x(\tau)h(t-\tau) = e^{\tau} e^{-(t-\tau)} = e^{-t} e^{2\tau}.$$

Case 1: for $t < 0$, the overlap ends at $\tau = t$

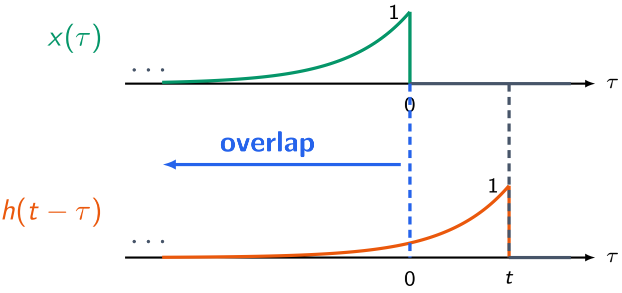
Overlap: $-\infty \quad < \quad \tau \quad \leq \quad t$



$$y(t) = \int_{-\infty}^{t} e^{\tau} e^{-(t-\tau)}\, d\tau = e^{-t} \int_{-\infty}^{t} e^{2\tau}\, d\tau = \frac{1}{2} e^{t}, \qquad t < 0.$$

Case 2: for $t \geq 0$, the overlap ends at $\tau = 0$

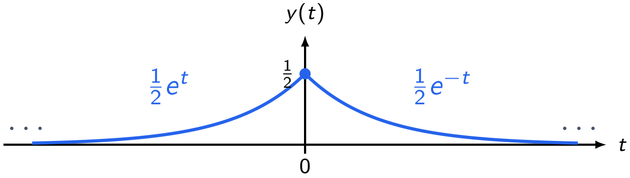
Overlap: $-\infty < \tau \leq 0$



$$y(t) = \int_{-\infty}^{0} e^{\tau} e^{-(t-\tau)}\, d\tau = e^{-t} \int_{-\infty}^{0} e^{2\tau}\, d\tau = \frac{1}{2} e^{-t}, \qquad t \geq 0.$$

Final answer: a two-sided decaying exponential

$$y(t) = \begin{cases} \dfrac{1}{2}e^{t}, & t < 0, \\ \dfrac{1}{2}e^{-t}, & t \geq 0. \end{cases}$$



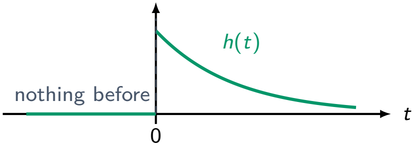
Causality and Stability

Causality: No Time Travel Please!!!

For an LTI system, causality is visible from $h(t)$

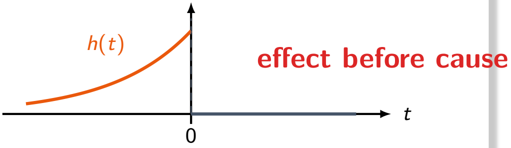
The impulse response $h(t)$ shows what the system does after we poke it with $\delta(t)$ at $t = 0$.

Causal



$$h(t) = 0 \quad \text{for } t < 0.$$

Noncausal



$$h(t) \neq 0 \quad \text{for some } t < 0.$$

Testing causality: the impulse response must be zero before 0

For continuous-time LTI systems

$$\boxed{\text{causal} \quad \Longleftrightarrow \quad h(t) = 0 \;\; \text{for all } t < 0}$$

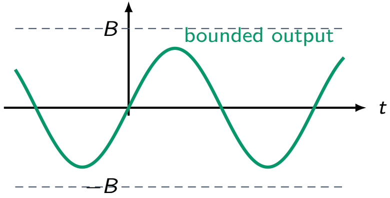
Discrete-time version

$$\boxed{\text{causal} \quad \Longleftrightarrow \quad h[n] = 0 \;\; \text{for all } n < 0}$$

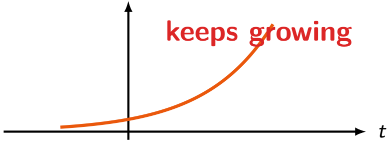
Stability means bounded input gives bounded output

A stable system does not turn a bounded input (value $\leq \; B$ always) into an output that shoots up to infinity and beyond.

Stable behavior



Unstable behavior



Stability depends on the size of the impulse response!

Intuition in continuous case

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)\, d\tau, \quad \text{yes, we shuffled it!}$$

If the input is bounded, say

$$|x(t)| \leq B \quad \text{for all } t, \text{ then, } |x(t-\tau)| \leq B.$$

$$|y(t)| \leq B \int_{-\infty}^{\infty} |h(\tau)|\, d\tau.$$

Stable LTI system: The total area under $|h(t)|$ must be finite

Continuous-time LTI stability test

$$\boxed{\text{stable} \quad \Longleftrightarrow \quad \int_{-\infty}^{\infty} |h(t)|\, dt < \infty}$$

Discrete-time version

$$\boxed{\text{stable} \quad \Longleftrightarrow \quad \sum_{n=-\infty}^{\infty} |h[n]| < \infty}$$

Two simple little problems

Assume an **LTI system** with impulse response $h(t)$.

1. If the system is **causal**, is it possible for $h(t)$ to be periodic?

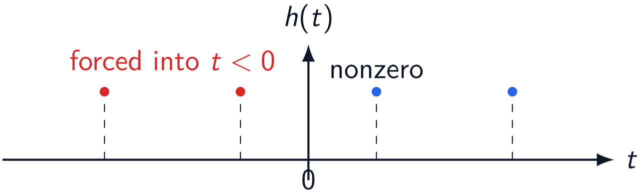
2. If the system is **stable**, is it possible for $h(t)$ to be periodic?

Answer 1: Causality vs Periodicity

If $h(t)$ is periodic and nonzero somewhere for $t > 0$, then the same nonzero value repeats backward:

$$h(t_0) = h(t_0 - T) = h(t_0 - 2T) = \cdots$$

Eventually, $t_0 - kT < 0$, which contradicts causality.

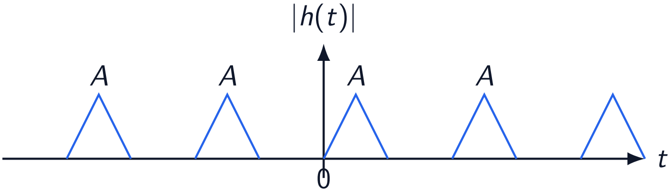


Therefore, only the trivial impulse response $h(t) = 0$ can be both causal and periodic.

Answer 2: Stability vs Periodicity

A nonzero periodic impulse response repeats forever. So if one period has positive area A under one period, then over infinitely many periods,

$$\int_{-\infty}^{\infty} |h(t)|\, dt = A + A + A + \cdots = \infty.$$



Therefore, only the trivial impulse response $h(t) = 0$ can be both stable and periodic.

Practice: decide causality and stability

For each impulse response, decide whether the LTI system is causal and stable.

System 1

$$h_1(t) = e^{2t}u(-2-t)$$

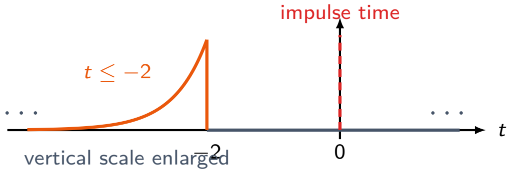
System 2

$$h_2(t) = te^{-t}u(t)$$

Answer 1: $h_1(t) = e^{2t}u(-2-t)$ is not causal

Support of $h_1(t)$

$$u(-2-t) = 1 \iff -2 - t \geq 0 \iff t \leq -2.$$



$h_1(t)$ is nonzero before $t = 0$. So the system is **not causal**.

Answer 1: $h_1(t) = e^{2t}u(-2-t)$ is stable

Stability test

$$\int_{-\infty}^{\infty} |h_1(t)|\, dt = \int_{-\infty}^{-2} e^{2t}\, dt.$$

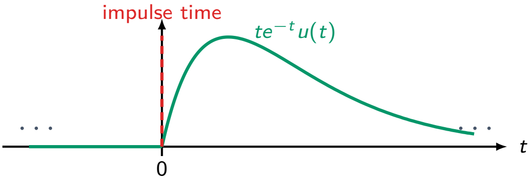
$$\int_{-\infty}^{-2} e^{2t}\, dt = \left[\frac{1}{2}e^{2t}\right]_{-\infty}^{-2} = \frac{1}{2}e^{-4} < \infty.$$

Therefore, the system is **stable**.

Answer 2: $h_2(t) = te^{-t}u(t)$ is causal

Support of $h_2(t)$

$$u(t) = 0 \quad \text{for } t < 0.$$



$h_2(t)$ is zero before $t = 0$. So the system is **causal**.

Answer 2: $h_2(t) = te^{-t}u(t)$ is stable

Stability test

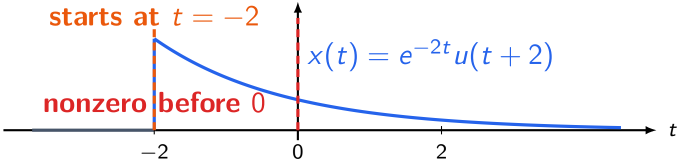
$$\int_{-\infty}^{\infty} |h_2(t)|\, dt = \int_0^{\infty} te^{-t}\, dt.$$

$$\int_0^{\infty} te^{-t}\, dt = 1 < \infty.$$

Therefore, the system is **stable**.

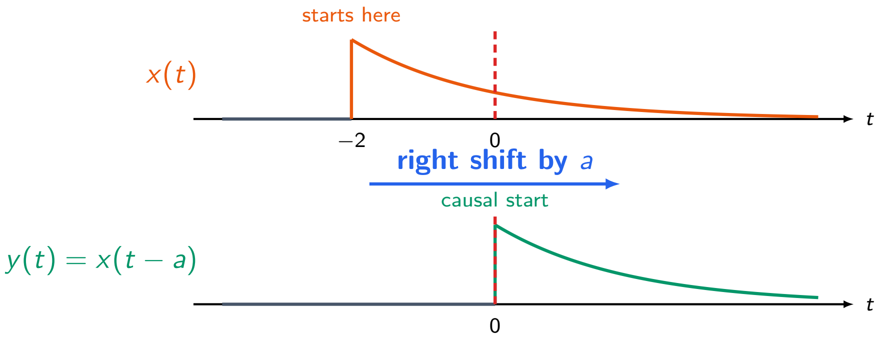
Problem: how much delay makes it causal?

Let $x(t) = e^{-2t}u(t+2)$, and $y(t) = x(t-a)$.
Find the minimum value of a so that $y(t)$ is causal.



How far must we shift this signal to the right?

Answer: shift it right by at least 2



For causality, $-2 + a \geq 0$, so $\boxed{a \geq 2.}$

Thank you!